

LABORATORY WORK 3

DML

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PostgreSQL 16 - airport_db

15 tasks with one full-page code-and-result screenshot per task

Task 1

Generate and insert 200 random airport rows

SQL CODE

Query Tool · task_01.sql Executed

```
WITH base AS (  
    SELECT COALESCE(MAX(airport_id), 0) AS max_id FROM airport  
) , inserted AS (  
    INSERT INTO airport  
        (airport_id, airport_name, country, state, city, created_at, updated_at)  
    SELECT base.max_id + g,  
        'Generated Airport ' || g || '-' || substr(md5(random()::text), 1, 6),  
        (ARRAY['Kazakhstan','Turkey','France','Brazil','Poland'])  
        [1 + floor(random() * 5)::int],  
        (ARRAY['North','South','East','West','Central'])  
        [1 + floor(random() * 5)::int],  
        'City ' || g,  
        CURRENT_TIMESTAMP, CURRENT_TIMESTAMP  
    FROM base CROSS JOIN generate_series(1, 200) AS g  
    RETURNING airport_id  
)  
SELECT COUNT(*) AS inserted_rows,  
    MIN(airport_id) AS first_id,  
    MAX(airport_id) AS last_id  
FROM inserted;
```

DATA OUTPUT

Result			INSERT 0 200
inserted_rows	first_id	last_id	
200	7	206	

Task 2

Add KazAir based in Kazakhstan

SQL CODE

Query Tool · task_02.sql

Executed

```
WITH next_id AS (  
    SELECT COALESCE(MAX(airline_id), 0) + 1 AS airline_id FROM airline  
)  
INSERT INTO airline  
    (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)  
SELECT airline_id, 'KZA', 'KazAir', 'Kazakhstan',  
    CURRENT_TIMESTAMP, CURRENT_TIMESTAMP  
FROM next_id  
RETURNING airline_id, airline_code, airline_name, airline_country;
```

DATA OUTPUT

Result				INSERT 0 1
airline_id	airline_code	airline_name	airline_country	
6	KZA	KazAir	Kazakhstan	

Task 3

Change the KazAir country to Turkey

SQL CODE

Query Tool · task_03.sql

Executed

```
UPDATE airline
SET airline_country = 'Turkey',
    updated_at = CURRENT_TIMESTAMP
WHERE airline_name = 'KazAir'
RETURNING airline_id, airline_name, airline_country, updated_at;
```

DATA OUTPUT

Result			UPDATE 1
airline_id	airline_name	airline_country	updated_at
6	KazAir	Turkey	2026-09-28 21:40:00

Task 4

Insert AirEasy, FlyHigh and FlyFly

SQL CODE

Query Tool · task_04.sql

Executed

```
WITH base AS (  
    SELECT COALESCE(MAX(airline_id), 0) AS max_id FROM airline  
) , new_airlines(offset_id, airline_code, airline_name, airline_country) AS (  
    VALUES (1, 'AE', 'AirEasy', 'France'),  
            (2, 'FH', 'FlyHigh', 'Brazil'),  
            (3, 'FF', 'FlyFly', 'Poland')  
)  
INSERT INTO airline  
    (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)  
SELECT base.max_id + offset_id, airline_code, airline_name, airline_country,  
       CURRENT_TIMESTAMP, CURRENT_TIMESTAMP  
FROM base CROSS JOIN new_airlines  
RETURNING airline_id, airline_code, airline_name, airline_country;
```

DATA OUTPUT

Result				INSERT 0 3
airline_id	airline_code	airline_name	airline_country	
7	AE	AirEasy	France	
8	FH	FlyHigh	Brazil	
9	FF	FlyFly	Poland	

Task 5

Delete flights arriving in 2024

SQL CODE

Query Tool · task_05.sql

Executed

```
DELETE FROM flights
WHERE sch_arrival_time >= DATE '2024-01-01'
AND sch_arrival_time < DATE '2025-01-01'
RETURNING flight_id, sch_arrival_time;
```

DATA OUTPUT

Result		DELETE 2
flight_id	sch_arrival_time	
201	2024-03-12 10:15:00	
202	2024-11-08 18:30:00	

Task 6

Increase every ticket price by 15 percent

SQL CODE

Query Tool · task_06.sql

Executed

```
WITH updated AS (  
  UPDATE booking  
  SET ticket_price = ROUND(ticket_price * 1.15, 2),  
      updated_at = CURRENT_TIMESTAMP  
  RETURNING ticket_price  
)  
SELECT COUNT(*) AS updated_rows,  
       MIN(ticket_price) AS minimum_price,  
       MAX(ticket_price) AS maximum_price  
FROM updated;
```

DATA OUTPUT

Result			UPDATE 6
updated_rows	minimum_price	maximum_price	
6	9200.00	212750.00	

Task 7

Delete tickets priced below 10000

SQL CODE

Query Tool · task_07.sql

Executed

```
DELETE FROM booking
WHERE ticket_price < 10000
RETURNING booking_id, flight_id, passenger_id, ticket_price;
```

DATA OUTPUT

Result				DELETE 1
booking_id	flight_id	passenger_id	ticket_price	
1006	101	1	9200.00	

Task 8

Replace missing airline codes with UNK

SQL CODE

Query Tool · task_08.sql

Executed

```
UPDATE airline
SET airline_code = 'UNK',
    updated_at = CURRENT_TIMESTAMP
WHERE airline_code IS NULL
RETURNING airline_id, airline_code, airline_name, airline_country;
```

DATA OUTPUT

Result			UPDATE 1
airline_id	airline_code	airline_name	airline_country
5	UNK	Legacy Air	Unknown

Task 9

Delete old unchecked baggage checks

SQL CODE

Query Tool · task_09.sql

Executed

```
DELETE FROM baggage_check
WHERE created_at < TIMESTAMP '2023-06-01 00:00:00'
  AND check_result = 'Not checked'
RETURNING baggage_check_id, check_result, created_at;
```

DATA OUTPUT

Result			DELETE 1
baggage_check_id	check_result	created_at	
1	Not checked	2023-05-10 08:00:00	

Task 10

Delete airports with missing state in Mlawe or Kepuh

SQL CODE

Query Tool · task_10.sql

Executed

```
DELETE FROM airport
WHERE state IS NULL
  AND city IN ('Mlawe', 'Kepuh')
RETURNING airport_id, airport_name, city, state;
```

DATA OUTPUT

Result			DELETE 2
airport_id	airport_name	city	state
5	Mlawe Regional Airport	Mlawe	NULL
6	Kepuh Regional Airport	Kepuh	NULL

Task 11

Insert an unchecked baggage inspection and return its ID

SQL CODE

Query Tool · task_11.sql

Executed

```
WITH selected_booking AS (  
    SELECT booking_id, passenger_id  
    FROM booking  
    ORDER BY booking_id  
    LIMIT 1  
) , next_id AS (  
    SELECT COALESCE(MAX(baggage_check_id), 0) + 1 AS baggage_check_id  
    FROM baggage_check  
)  
INSERT INTO baggage_check  
    (baggage_check_id, check_result, created_at, updated_at, booking_id, passenger_id)  
SELECT baggage_check_id, 'Not checked', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP,  
    booking_id, passenger_id  
FROM next_id CROSS JOIN selected_booking  
RETURNING baggage_check_id, created_at;
```

DATA OUTPUT

Result		INSERT 0 1
baggage_check_id	created_at	
6	2026-09-28 21:40:11	

Task 12

Convert all airline countries to uppercase

SQL CODE

Query Tool · task_12.sql

Executed

```
WITH updated AS (  
  UPDATE airline  
    SET airline_country = UPPER(airline_country),  
        updated_at = CURRENT_TIMESTAMP  
  WHERE airline_country <> UPPER(airline_country)  
  RETURNING airline_country  
)  
SELECT COUNT(*) AS updated_rows,  
       COUNT(*) FILTER (WHERE airline_country = UPPER(airline_country))  
       AS uppercase_rows  
FROM updated;
```

DATA OUTPUT

Result		UPDATE 8
updated_rows	uppercase_rows	
8	8	

Task 13

Update airline 5 and its timestamp

SQL CODE

Query Tool · task_13.sql

Executed

```
UPDATE airline
SET airline_name = 'Global Airways',
    airline_country = 'United Kingdom',
    updated_at = CURRENT_DATE
WHERE airline_id = 5
RETURNING airline_id, airline_name, airline_country, updated_at;
```

DATA OUTPUT

Result				UPDATE 1
airline_id	airline_name	airline_country	updated_at	
5	Global Airways	United Kingdom	2026-09-28 00:00:00	

Task 14

Set Capital District for three airport cities

SQL CODE

Query Tool · task_14.sql

Executed

```
UPDATE airport
SET state = 'Capital District',
    updated_at = CURRENT_TIMESTAMP
WHERE city IN ('Astana', 'London', 'Tokyo')
RETURNING airport_id, airport_name, city, state;
```

DATA OUTPUT

Result			UPDATE 3
airport_id	airport_name	city	state
2	Nursultan Nazarbayev International Airport	Astana	Capital District
3	Heathrow Airport	London	Capital District
4	Haneda Airport	Tokyo	Capital District

Task 15

Complete unchecked baggage checks from March 2024

SQL CODE

Query Tool · task_15.sqlExecuted

```
UPDATE baggage_check
SET check_result = 'Checked',
    updated_at = CURRENT_TIMESTAMP
WHERE created_at >= TIMESTAMP '2024-03-01 00:00:00'
    AND created_at <  TIMESTAMP '2024-04-01 00:00:00'
    AND check_result = 'Not checked'
RETURNING baggage_check_id, check_result, created_at, updated_at;
```

DATA OUTPUT

Result			UPDATE 2
baggage_check_id	check_result	created_at	updated_at
3	Checked	2024-03-05 10:00:00	2026-09-28 21:40:00
4	Checked	2024-03-20 11:00:00	2026-09-28 21:40:00